

# AN AUTOMATED STOCHASTIC VOLATILITY MODEL PARAMETER CALIBRATION ROUTINE WITH OPTIONAL JUMPS

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**ABSTRACT.** We specify a routine that calibrates the five canonical Heston [1993] parameters  $\Phi = (\theta, \kappa, \eta, \rho, v_0)$  to a day of CBOE S&P 500 (SPX) intraday option trades. The routine performs *one* calibration per trading day over a pooled, moneyness-normalised implied-volatility surface, and is framed throughout as a box-constrained nonlinear least-squares problem solved from multiple seeds. We make explicit every modelling choice: the volume-weighted reference spot  $S_{\text{ref}}$ , the moneyness-normalised and grid-snapped strike  $K^*$ , the surface-coverage requirements, the configurable in-engine objective (defaulting to implied-volatility “IV-space” residuals), the IV-space restart selection and acceptance gate, and the boundary-pegging rejection rule. The same routine optionally fits the Bates [1996] extension, which adds Merton lognormal jumps to the Heston dynamics, selected by a run-time flag and calibrated over the identical surface and gate. We keep Heston as the default throughout and present Bates as the eight-parameter generalisation it specialises from.

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## 1. The model

**1.1. Basis.** Under the risk-neutral measure, with constant risk-free rate  $r$  and continuous dividend yield  $g$ , the log-price  $X_t = \ln S_t$  and instantaneous variance  $v_t$  of the SPX follow the Heston [1993] dynamics

$$dX_t = (r - g - \frac{1}{2}v_t) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t), \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \eta\sqrt{v_t} dW_t, \quad (2)$$

where  $W, B$  are independent Brownian motions, so  $\text{corr}(dX_t, dv_t) = \rho dt$ . The parameter vector is

$$\Phi = (\theta, \kappa, \eta, \rho, v_0) \in \mathbb{R}^5, \quad (3)$$

with  $\theta > 0$  the long-run variance,  $\kappa > 0$  the mean-reversion speed,  $\eta > 0$  the volatility of variance (QuantLib's `sigma`),  $\rho \in [-1, 1]$  the leverage correlation, and  $v_0 > 0$  the initial variance. The Feller quantity [Feller, 1951]

$$\mathcal{F}(\Phi) = 2\kappa\theta - \eta^2 \quad (4)$$

is reported per fit;  $\mathcal{F}(\Phi) \geq 0$  guarantees  $v_t > 0$  almost surely. It is not imposed as a constraint (short-tenor SPX fits routinely violate it without being otherwise unreasonable; see §6).

**1.2. Jumps.** The Bates [1996] model extends Heston with Merton [1976] log-normal jumps in the log-price. The variance dynamics (2) are unchanged. The level equation (1) gains a compound-Poisson jump term:

$$dX_t = (r - g - \frac{1}{2}v_t - \lambda\xi) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t) + dJ_t. \quad (5)$$

Here  $J_t = \sum_{n=1}^{N_t} Y_n$  is a compound-Poisson process:  $N_t$  is a Poisson process of intensity  $\lambda \geq 0$  (expected jumps per year), independent of  $W$  and  $B$ , and the log-jump sizes  $Y_n \sim \mathcal{N}(\nu, \delta^2)$  are i.i.d. The compensator

$$\xi = \mathbb{E}[e^Y - 1] = e^{\nu + \delta^2/2} - 1 \quad (6)$$

offsets the jump drift so the discounted price stays a martingale, leaving the level drift  $r - g$  in expectation. The augmented parameter vector is

$$\Theta = (\theta, \kappa, \eta, \rho, v_0, \lambda, \nu, \delta) \in \mathbb{R}^8, \quad (7)$$

the five Heston parameters of (1)–(2) plus jump intensity  $\lambda \geq 0$ , mean log-jump  $\nu$ , and log-jump volatility  $\delta > 0$ . At  $\lambda = 0$  the jump term vanishes and (5) collapses to the Heston level (1), so Heston is the  $\lambda = 0$  face of the Bates parameter space. The model is selected at run time by a flag; the Heston fit is the default. The Feller quantity (4) is built from the diffusive parameters alone, so jumps do not enter it; it is reported under both models.

**1.3. Pricing operators.** Fix a valuation with maturity  $\tau > 0$  (in years), strike  $K$ , and type  $w \in \{+1, -1\}$  (+1 call, -1 put). Write the forward and discount factor under flat curves as

$$F(S, \tau) = S e^{(r-g)\tau}, \quad D(\tau) = e^{-r\tau}. \quad (8)$$

**Black price** [Black, 1976, Black and Scholes, 1973]. With total standard deviation  $\Sigma = \sigma\sqrt{\tau}$ ,

$$C_B(\sigma; S, K, \tau, w) = w D(\tau) \left[ F \mathcal{N}(wd_+) - K \mathcal{N}(wd_-) \right], \quad (9)$$

$$d_{\pm} = \frac{\ln(F/K)}{\Sigma} \pm \frac{1}{2}\Sigma,$$

where  $\mathcal{N}$  is the standard normal CDF and  $F = F(S, \tau)$ .

**Heston price.** The Heston model price  $C_H(\Phi; S, K, \tau, w)$  is the semi-analytic value

$$C_H(\Phi; S, K, \tau, +1) = S e^{-g\tau} P_1 - K e^{-r\tau} P_2, \quad (10)$$

$$C_H(\cdot, -1) = C_H(\cdot, +1) - S e^{-g\tau} + K e^{-r\tau},$$

with the in-the-money probabilities, for  $j \in \{1, 2\}$ ,

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[ \frac{e^{-iu \ln K} f_j(u; \Phi, \tau, \ln F)}{iu} \right] du, \quad (11)$$

and characteristic functions  $f_j = \exp(C_j + D_j v_0 + iu \ln F)$  given by the standard coefficients

$$C_j = \frac{\kappa\theta}{\eta^2} \left[ (b_j - \rho\eta iu + d_j)\tau - 2 \ln \frac{1 - \gamma_j e^{d_j\tau}}{1 - \gamma_j} \right], \quad (12)$$

$$D_j = \frac{b_j - \rho\eta iu + d_j}{\eta^2} \frac{1 - e^{d_j\tau}}{1 - \gamma_j e^{d_j\tau}},$$

$$d_j = \sqrt{(\rho\eta iu - b_j)^2 - \eta^2(2\mu_j iu - u^2)}, \quad \gamma_j = \frac{b_j - \rho\eta iu + d_j}{b_j - \rho\eta iu - d_j}, \quad (13)$$

where  $\mu_1 = \frac{1}{2}$ ,  $\mu_2 = -\frac{1}{2}$ ,  $b_1 = \kappa - \rho\eta$ ,  $b_2 = \kappa$ . In code this integral is evaluated by the QuantLib [Ballabio and the QuantLib contributors] **AnalyticHestonEngine** (which uses a branch-stable formulation of (11)); equations (10)–(11) define the operator it realises.

**Bates price.** The Bates price  $C_{\text{Bates}}(\Theta; S, K, \tau, w)$  has the same form as the Heston price (10)–(11), with each characteristic function  $f_j$  multiplied by the Merton lognormal jump factor of §1.2:

$$f_j^B(u) = f_j(u) e^{\tau \psi_j(u)}, \quad \psi_j(u) = \lambda \left( e^{iu_j \nu - \frac{1}{2} \delta^2 u_j^2} - 1 \right) - iu_j \lambda \xi, \quad (14)$$

with  $u_2 = u$ ,  $u_1 = u - i$  (the same measure shift that distinguishes  $f_1$  from  $f_2$ ), and  $\xi = e^{\nu + \delta^2/2} - 1$  the compensator (6). The exponent  $\psi_j$  is the Merton log-jump cumulant, compensated by  $-iu_j \lambda \xi$  so the drift correction of (5) carries through. At  $\lambda = 0$  the factor is 1 and  $C_{\text{Bates}}$  reduces to  $C_H$ . In code the integral is evaluated by the QuantLib **BatesEngine**, the jump-augmented counterpart of the **AnalyticHestonEngine**.

**Implied volatility.** The Black-implied volatility of a price  $C$  is the inverse of (9) in its first argument,

$$\mathcal{I}(C; S, K, \tau, w) = \{ \sigma > 0 : C_B(\sigma; S, K, \tau, w) = C \}, \quad (15)$$

computed numerically (QuantLib **blackFormulaImpliedStdDev** on the forward (8), returning  $\Sigma$ , then  $\sigma = \Sigma/\sqrt{\tau}$ ). This operator defines the IV-space residuals of §5.

## 2. Data and per-day notation

The pipeline ingests CBOE trade-level files, one per trading day  $d$ , restricted upstream to out-of-the-money (OTM) quotes. A day is a set of trade records indexed by  $i$ , each carrying

$$(t_i, K_i, w_i, q_i, P_i, \sigma_i, S_i, \tau_i), \quad (16)$$

being respectively the quote timestamp, strike, type  $w_i \in \{+1, -1\}$ , traded size, trade price, trade-implied volatility, the underlying level *at trade time*  $t_i$ , and the days to maturity. Maturities enter the engine as  $\tau_i$  in years under Actual/365, i.e.  $T_i = \tau_i/365$ .

The day is rate-stamped with  $r = r(d)$  and  $g = g(d)$ , the most recent risk-free and dividend rates on or before  $d$  (a flat-forward curve for each; under flat curves the evaluation date is immaterial to  $T_i$ , so it is set to the QuantLib “today”).

**Pre-filter.** We keep only economically usable, Heston-appropriate quotes. Write the ratio moneyness  $\varrho_i = e^{-|\ln(K_i/S_i)|} \in (0, 1]$  (it equals 1 at the money and falls as the strike moves out). The kept set is

$$\mathcal{D} = \{i : \sigma_i > 0, \tau_{\min} \leq \tau_i \leq \tau_{\max}, \underline{m} < \varrho_i < \bar{m}\}. \quad (17)$$

The maturity floor  $\tau_{\min}$  (Table 1) drops ultra-short maturities, which the diffusive Heston smile fits poorly and which otherwise drive  $\eta, \kappa$  to Feller-violating extremes that pollute the pooled fit; the ceiling  $\tau_{\max}$  drops thin, stale long tenors. The cutoff  $\bar{m}$  removes the near-ATM rows, and the floor  $\underline{m}$  removes the deep-OTM lottery-ticket tail ( $\varrho_i < \underline{m}$ , roughly  $|\ln(K_i/S_i)| > 0.51$ ), whose extreme prices peg the fit to its bounds. Flooring that tail recovers acceptance and tightens the fit while keeping the full tradeable wing.

## 3. Handling intraday spot movement

A Heston calibration is anchored to a *single* spot  $S$ , yet the trades in  $\mathcal{D}$  occur across an intraday range of underlying levels  $\{S_i\}$ . We reconcile the two with one reference spot and a moneyness re-striking.

**Reference spot.** Define the volume-weighted reference spot

$$S_{\text{ref}} = \frac{\sum_{i \in \mathcal{D}} q_i S_i}{\sum_{i \in \mathcal{D}} q_i}. \quad (18)$$

The intraday range and its flag are

$$R = \frac{\max_i S_i}{\min_i S_i} - 1, \quad \text{HIGH\_MOVE} = \mathbf{1}\{R > R_{\max}\}. \quad (19)$$

A flagged day ( $R > R_{\max}$ ) is still calibrated and written, but the normalisation below is recorded as strained (its  $S_{\text{ref}}$  should be treated with suspicion).

**Moneyness normalisation and grid snap.** Each trade retains its moneyness at trade time and is re-struck to the reference spot, then snapped to the SPX strike grid of width  $\Delta$ :

$$m_i = \frac{K_i}{S_i}, \quad K_i^* = \Delta \cdot \text{round}\left(\frac{m_i S_{\text{ref}}}{\Delta}\right). \quad (20)$$

The rationale is sticky-moneyness: over a session the implied-volatility smile is approximately stationary in  $K/S$  rather than in the absolute strike. Re-striking by (20) therefore re-centres the whole day onto  $S_{\text{ref}}$  so that trades executed at

different  $S_i$  but the *same* moneyness share a common surface column. The grid snap collapses the continuum of  $K_i^*$  values (one per trade) back onto a clean strike lattice. Pricing diagnostics (§7) nonetheless use each contract’s *original*  $(S_i, K_i)$ , since the calibrated parameters are spot-independent.

#### 4. Surface construction

From the normalised trades we build a partial  $K^* \times \tau$  implied-volatility surface, selecting the most liquid quotes nearest the money. The trades entering this stage are already the pre-filtered set  $\mathcal{D}$ , so every candidate strike sits in the OTM moneyness band  $\underline{m} < \varrho_i < \bar{m}$  of (17). That band fixes the reach of each wing: the cutoff  $\bar{m}$  sets the inner edge, the strikes closest to the money, and the floor  $\underline{m}$  sets the outer edge, where the deep-OTM tail is dropped. “Nearest the money” below therefore means just inside  $\bar{m}$ , not at the money itself.

**Maturity selection.** A maturity enters the surface only if it carries both wings. For each distinct maturity  $\tau$  present in  $\mathcal{D}$ , let  $n_c(\tau)$  and  $n_p(\tau)$  be the number of distinct  $K^*$  on its call and put wings, and let  $V(\tau) = \sum_{i \in \mathcal{D}: \tau_i = \tau} q_i$  be its traded volume. Call  $\tau$  *admissible* when each wing carries at least  $\underline{n}_k$  distinct strikes. Keep the  $\bar{N}_t$  highest-volume admissible maturities,

$$\mathcal{T} = \text{top-}\bar{N}_t \text{ by } V(\tau) \text{ among } \{\tau : n_c(\tau) \geq \underline{n}_k, n_p(\tau) \geq \underline{n}_k\}. \quad (21)$$

The wing test precedes the volume cap. A maturity with a one-sided or single-strike wing is discarded before the cap, rather than consuming one of its  $\bar{N}_t$  slots. The kept set falls below  $\bar{N}_t$  only when fewer than  $\bar{N}_t$  maturities are admissible.

**Strike selection (per maturity, per wing).** Within each kept maturity  $\tau \in \mathcal{T}$ , separate calls ( $w_i = +1$ , OTM with  $K_i^* \gtrsim S_{\text{ref}}$ ) from puts ( $w_i = -1$ , OTM with  $K_i^* \lesssim S_{\text{ref}}$ ) and keep the  $\bar{n}_k$  nearest-the-money distinct strikes on each wing:

$$\mathcal{K}_\tau = \underbrace{\{\bar{n}_k \text{ largest } K^* \text{ among puts at } \tau\}}_{\text{put wing}} \cup \underbrace{\{\bar{n}_k \text{ smallest } K^* \text{ among calls at } \tau\}}_{\text{call wing}}. \quad (22)$$

The  $\bar{n}_k$  cap binds only where a wing carries more than  $\bar{n}_k$  distinct strikes inside the band; otherwise the wing’s outer edge is set by the floor  $\underline{m}$  alone.

**Pivot to a surface.** For each retained cell  $(k, \tau)$  with  $k \in \mathcal{K}_\tau$ , set the surface entry to the implied volatility of the *highest-volume* trade in that cell, the one with the largest traded size,

$$\hat{\sigma}_{k,\tau} = \sigma_{i^*}, \quad i^* = \arg \max_{i: K_i^* = k, \tau_i = \tau} q_i, \quad (23)$$

and leave it missing otherwise. When several trades tie at the maximum size the pick is arbitrary. Let

$$\mathcal{N} = \{(k, \tau) : \hat{\sigma}_{k,\tau} \text{ defined}\} \quad (24)$$

be the set of populated cells; these become the calibration targets (helpers).

**Coverage requirement.** To keep the five-parameter fit identified we require a genuinely two-dimensional surface:

$$|\{\tau : (\cdot, \tau) \in \mathcal{N}\}| \geq M_t, \quad |\{k : (k, \cdot) \in \mathcal{N}\}| \geq M_k, \quad |\mathcal{N}| \geq M_{\text{cell}}. \quad (25)$$

A day failing (25) (or with no usable maturities) is dropped: it contributes no calibration and clears its stale diagnostics.

### 5. The optimization problem

The routine calibrates one parameter vector per day. Under the default Heston model this is  $\Phi$  of (3) with  $\dim \Phi = 5$ ; under Bates it is  $\Theta$  of (7) with  $\dim \Theta = 8$ . The two are nested. They share their first  $\dim \Phi$  coordinates, the diffusive parameters  $(\theta, \kappa, \eta, \rho, v_0)$ , and  $\Theta$  appends the jump triple  $(\lambda, \nu, \delta)$ . Setting  $\lambda = 0$  collapses the jump term (5) and recovers  $\Phi$ , so Heston is the  $\lambda = 0$  face of Bates. We write  $\Psi$  for the active vector, either  $\Phi$  or  $\Theta$ , and  $C^{\text{mod}}(\Psi; \cdot)$  for the active model price: the Heston price  $C_H$  (10) when  $\Psi = \Phi$ , and the Bates price  $C_{\text{Bates}}$  (14) when  $\Psi = \Theta$ . Everything below is written in  $\Psi$  and holds for either model. The only place the model enters is the box dimension  $\dim \Psi$  (26) and, through it, the bounds tables.

Let  $\mathcal{N}$  index the targets with market data  $\{(\hat{\sigma}_n, K_n^*, T_n, w_n)\}_{n \in \mathcal{N}}$ , all priced at the single spot  $S_{\text{ref}}$ . Define the feasible box

$$\mathcal{B} = \prod_{j=1}^{\dim \Psi} [\ell_j, h_j] \subset \mathbb{R}^{\dim \Psi}, \quad (26)$$

with  $(\ell, h)$  in the `model.params()` order of Table 2: the five Heston coordinates, extended by the three jump bounds when  $\Psi = \Theta$ . The  $\lambda$  floor is exactly 0, so the Bates box contains the Heston model on its  $\lambda = 0$  face.

**5.1. Residuals.** Two residual systems are defined: an IV-space system and a price-space system. The in-engine objective is configurable. It defaults to the IV-space system (30) below, and can instead be set to the price-space system (27). The choice changes only what each restart minimises. Selection (34) and the acceptance gate (36) always run on the IV-space error  $J_{\text{iv}}$ , so under the default objective the optimizer minimises the very error that then selects and gates.

**Price-space (alternative in-engine objective).** For each target the market value is the Black price at the quoted vol,  $C_n^{\text{mkt}} = C_B(\hat{\sigma}_n; S_{\text{ref}}, K_n^*, T_n, w_n)$ , and the relative-price residual is

$$\varepsilon_n^{\text{px}}(\Psi) = \frac{C^{\text{mod}}(\Psi; S_{\text{ref}}, K_n^*, T_n, w_n) - C_n^{\text{mkt}}}{C_n^{\text{mkt}}}. \quad (27)$$

The associated objective is the root-mean-square

$$J_{\text{px}}(\Psi) = \left( \frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{px}}(\Psi)^2 \right)^{1/2}. \quad (28)$$

The per-target residual  $\varepsilon_n^{\text{px}}$  is the (relative-price) error each QuantLib helper exposes. Levenberg–Marquardt [Marquardt, 1963] is a sum-of-squares method, so it drives the squared residual norm  $\sum_n (\varepsilon_n^{\text{px}})^2 = |\mathcal{N}| J_{\text{px}}(\Psi)^2$  to a local minimum (equivalently  $J_{\text{px}}$ , with which it shares a minimiser).

**IV-space (default in-engine objective; selection and gate).** Invert each model price back to a Black vol via (15) and compare to the market vol:

$$\begin{aligned} \sigma_n^{\text{mod}}(\Psi) &= \mathcal{I}(C^{\text{mod}}(\Psi; S_{\text{ref}}, K_n^*, T_n, w_n); S_{\text{ref}}, K_n^*, T_n, w_n), \\ \varepsilon_n^{\text{iv}}(\Psi) &= \sigma_n^{\text{mod}}(\Psi) - \hat{\sigma}_n, \end{aligned} \quad (29)$$

with objective

$$J_{\text{iv}}(\Psi) = \left( \frac{1}{|\mathcal{N}'|} \sum_{n \in \mathcal{N}'} \varepsilon_n^{\text{iv}}(\Psi)^2 \right)^{1/2}, \quad \mathcal{N}' = \{n : \sigma_n^{\text{mod}}(\Psi) \text{ inverts finitely}\}. \quad (30)$$

$J_{\text{iv}}$  is measured in volatility points; it is the natural surface-fit metric and, unlike  $J_{\text{px}}$ , is not inflated by cheap deep-OTM wings (a unit price error on a near-worthless contract is a large *relative* price error but a small vol error).

**5.2. Multi-start local optimization.** A single start into a non-convex problem is fragile [Mikhailov and Nögel, 2003], so we restart from a small, data-seeded set. Let the surface variance level be

$$\hat{v} = \text{clip}\left(\left(\text{median}_{(k,\tau) \in \mathcal{N}} \hat{\sigma}_{k,\tau}\right)^2, [10^{-3}, 0.25]\right), \quad (31)$$

and define the seed set  $\mathcal{S} = \{s_1, \dots, s_6\} \subset \mathcal{B}$  of  $(v_0, \kappa, \theta, \eta, \rho)$  tuples

$$\mathcal{S} = \begin{cases} (\hat{v}, 1.0, \hat{v}, 0.50, -0.70), & (\hat{v}, 3.0, \hat{v}, 1.00, -0.50), \\ (\hat{v}, 0.5, \hat{v}, 0.30, -0.90), & (\hat{v}, 5.0, \hat{v}, 0.80, -0.60), \\ (0.8\hat{v}, 2.0, 1.2\hat{v}, 0.60, -0.75), & (\hat{v}, 8.0, \hat{v}, 1.20, -0.40). \end{cases} \quad (32)$$

Under Bates each seed is extended by one fixed jump triple  $(\lambda, \nu, \delta) = (0.1, -0.1, 0.1)$ , so every restart starts from almost no jumps and grows them only if they tighten the fit; the restart count is unchanged. For each seed  $s \in \mathcal{S}$  we solve the box-constrained nonlinear least-squares problem

$$\hat{\Psi}_s = \arg \min_{\Psi \in \mathcal{B}} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{obj}}(\Psi)^2 \quad (\text{LM, started at } \Psi^{(0)} = s), \quad (33)$$

using Levenberg–Marquardt under the box constraint  $\mathcal{B}$  (`QuantLib NonhomogeneousBoundaryConstraint`). The per-target residual  $\varepsilon_n^{\text{obj}}$  is the configured in-engine objective: by default the IV-space residual  $\varepsilon_n^{\text{iv}}$  (29), optionally the price-space residual  $\varepsilon_n^{\text{px}}$  (27). The fit retained for the day is the restart with the smallest *IV-space* error,

$$\Psi^* = \arg \min_{s \in \mathcal{S}} J_{\text{iv}}(\hat{\Psi}_s). \quad (34)$$

Ranking by  $J_{\text{iv}}$  (not  $J_{\text{px}}$ ) removes the dependence on any single arbitrary guess and selects the economically best surface fit.

## 6. Acceptance gate

$\Psi^*$  is accepted only if it fits the surface tightly *and* lies in the interior of the box. Let the boundary-pegging predicate, with tolerance fraction  $b_{\text{tol}}$ , be

$$\text{PEGGED}(\Psi) = \bigvee_{j=1}^{\dim \Phi} \left[ (\Psi_j - \ell_j) < b_{\text{tol}} (h_j - \ell_j) \vee (h_j - \Psi_j) < b_{\text{tol}} (h_j - \ell_j) \right]. \quad (35)$$

The disjunction runs over the  $\dim \Phi$  diffusive coordinates for both models. Under Bates the three jump coordinates  $(\nu, \delta, \lambda)$  are exempt from the predicate:  $\lambda \rightarrow 0$  is a legitimate collapse to Heston rather than the optimizer hitting a wall, and when  $\lambda \approx 0$  the jump shape  $(\nu, \delta)$  is unidentified and may rest on a bound without meaning. The jump triple is still reported. The Feller quantity (4)

likewise stays the diffusive quantity under Bates and never gates. The day is accepted iff

$$\boxed{J_{\text{iv}}(\Psi^*) \leq \varepsilon_{\text{iv}} \quad \text{and} \quad \neg \text{PEGGED}(\Psi^*)}. \quad (36)$$

The first clause is the IV-space gate ( $\varepsilon_{\text{iv}} \approx 2$  vol points). The second rejects boundary fits: a parameter resting on a bound is the optimizer hitting a wall, not an identified value, so it is treated as a non-fit regardless of  $J_{\text{iv}}$ . The Feller quantity (4) is reported but never gates, since short-tenor SPX fits can violate it while remaining otherwise sound. A rejected or too-thin day yields no parameter row.

**Why an IV-space gate.** An earlier version gated on  $J_{\text{px}}$  directly. On a pooled day rich in deep-OTM wings,  $J_{\text{px}}$  is dominated by those cheap contracts and rejects fits that are excellent in vol points. Replacing the gate metric with  $J_{\text{iv}}$  aligns the acceptance criterion with the units the surface is quoted in; under it the genuine pooled-surface fit is  $\approx 0.8$ – $1.4$  vol points, and the residual rejections are boundary-pegged  $\kappa$  or  $\rho$  (a separate identification lever), not fit-quality failures.

## 7. Output and diagnostics

Every attempted day produces output. Results are routed by model and objective into one directory, `results/⟨model⟩/calibrations/⟨objective⟩/`, so a Heston run never shares a file with a Bates run, nor an IV-space run with a price-space one. That directory holds three artefacts: a parameter table of accepted days (`calibrations.csv`), a rejection table of dropped days (`rejections.csv`), and a folder of per-day repricing diagnostics (`calibration_tests/`). The two tables partition the attempted days. An accepted day writes one row to the parameter table and nothing to the rejection table; a rejected day does the reverse. So the accept rate and the rejection breakdown of §?? read straight off them.

**Accepted days: the parameter table.** An accepted day appends one row, keyed by the date  $d$ , to `calibrations.csv`. The row carries the rate-stamped inputs, the fitted vector, and fit and coverage diagnostics:

$$\begin{aligned} & (d; S_{\text{ref}}, r, g; \theta^*, \kappa^*, \rho^*, \eta^*, v_0^*; \mathcal{F}(\Psi^*); \\ & J_{\text{iv}}(\Psi^*), J_{\text{px}}(\Psi^*), |\mathcal{N}|; |\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|, \sum_i q_i; \\ & \min_i S_i, \max_i S_i, R, \text{HIGH\_MOVE}; t_{\text{calc}}). \end{aligned} \quad (37)$$

The blocks are: the reference spot (18) and the day’s rates; the fitted parameters  $\Psi^*$  with their Feller quantity (4); the two objective values (30) (28) and the helper count  $|\mathcal{N}|$ ; the surface coverage (kept maturities, distinct strikes, populated cells) and the day’s total traded volume; the intraday spot range with its flag (19); and  $t_{\text{calc}}$ , the latest quote timestamp on the surface. A boolean accept flag (always true in this file) is stored alongside. Under Bates the jump triple  $(\lambda^*, \nu^*, \delta^*)$  is inserted after  $v_0^*$  and nothing else changes. Because the calibration is once-daily, these rows accumulate into the one file, rewritten in date order at the end of each run.

**Rejected days: the rejection table.** A day dropped at any stage appends one row, also keyed by  $d$ , to `rejections.csv`, recording why:

$$(d; \text{REASON}, \text{DETAIL}; J_{\text{iv}}; |\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|). \quad (38)$$

The `REASON` is one of six categories, in pipeline order: `no_trades` (nothing survives the pre-filter (17)), `no_rate` (no  $r$  or  $g$  on or before  $d$ ), `thin` (the coverage requirement (25) fails), `pegged` (the fit clears the IV gate but a diffusive parameter is boundary-pegged (35)), `iv_miss` ( $J_{iv}(\Psi^*) > \varepsilon_{iv}$ ), and `no_fit` (no restart returned a finite fit). `DETAIL` is the human-readable string. The IV error and the coverage counts are filled in where the day reached the stage that computes them, and left blank otherwise: a `thin` day has coverage counts but no  $J_{iv}$ , a `no_trades` day has neither. The two tables together cover every attempted day, so the pegged-versus-thin-versus-miss split is auditable directly.

**Per-day repricing diagnostics.** Each accepted day also writes one file to `calibration_tests/`, holding the day’s repriced surface contracts. One representative trade per populated cell (the highest-volume one, matching the pivot (23)) is valued under  $\Psi^*$ . The valuation uses each contract’s *original* ( $S_i, K_i$ ), not ( $S_{ref}, K^*$ ): the structural parameters are spot-independent, and the honest comparison is against the actual trade. Each row records the original spot and strike, the snapped  $K^*$ , the fitted parameters, the market implied volatility  $\sigma_i$ , the Black price (9) at that vol, and the model price (the Heston price (10), or a Bates price column (14) under Bates).

## 8. Constants

| Symbol             | Code name             | Value     | Role                                    |
|--------------------|-----------------------|-----------|-----------------------------------------|
| $\tau_{min}$       | MIN_DTM               | 14        | minimum days to maturity                |
| $\tau_{max}$       | MAX_DTM               | 730       | maximum days to maturity                |
| $\underline{m}$    | OTM_MONEYNESSE_FLOOR  | 0.6       | minimum OTM ratio moneyness             |
| $\bar{m}$          | OTM_MONEYNESSE_CUTOFF | 0.98      | maximum OTM ratio moneyness             |
| $R_{max}$          | MAX_MOVE_PCT          | 0.03      | intraday-range flag threshold           |
| $\Delta$           | STRIKE_GRID           | 5.0       | SPX strike lattice for $K^*$ snap       |
| $\bar{N}_t$        | MAX_NT                | 20        | maturities kept (by volume)             |
| $\underline{n}_k$  | MIN_NK                | 2         | minimum distinct strikes per wing       |
| $\bar{n}_k$        | MAX_NK                | 40        | strikes kept per wing                   |
| $M_t$              | MIN_MATS              | 3         | minimum maturities in surface           |
| $M_k$              | MIN_STRIKES           | 5         | minimum strikes in surface              |
| $M_{cell}$         | MIN_CELLS             | 12        | minimum populated cells $ \mathcal{N} $ |
| $\varepsilon_{iv}$ | IV_RMSE_ACCEPT        | 0.02      | IV-space acceptance gate (vol pts)      |
| $b_{tol}$          | BOUND_TOL             | $10^{-3}$ | boundary-pegging tolerance fraction     |

TABLE 1. Surface, selection, and gate constants.

|              | $\theta$  | $\kappa$  | $\eta$    | $\rho$   | $v_0$     |              | $\lambda$ | $\nu$  | $\delta$  |
|--------------|-----------|-----------|-----------|----------|-----------|--------------|-----------|--------|-----------|
| lower $\ell$ | $10^{-4}$ | $10^{-2}$ | $10^{-2}$ | $-0.999$ | $10^{-4}$ | lower $\ell$ | 0.0       | $-0.5$ | $10^{-3}$ |
| upper $h$    | 1.0       | 20.0      | 2.0       | 0.5      | 1.0       | upper $h$    | 5.0       | 0.2    | 0.5       |

TABLE 2. Box bounds  $\mathcal{B}$  (26), in the `model.params()` order. Left: the five Heston coordinates ( $\theta, \kappa, \eta, \rho, v_0$ ); the  $\rho$  upper bound is kept slightly positive (equity leverage implies  $\rho < 0$ ) and  $\eta$  is capped at 2.0 (SPX vol-of-vol  $\sim 0.3$ – $1.2$ ). Right: the Bates jump bounds ( $\lambda, \nu, \delta$ ) appended when  $\Psi = \Theta$ ; the  $\lambda$  floor is exactly 0 (the fit may collapse to pure Heston) and  $\nu$  allows more negative room since equity jumps skew down.

## 9. Avenues for improvement

Rough volatility is the natural next paradigm, but a separate one. Empirical realised variance is well described by a fractional process with Hurst exponent  $H \approx 0.1$  [Gatheral et al., 2018], far from the  $H = \frac{1}{2}$  that the diffusive Heston and Bates dynamics assume. A rough model fits the steep short-maturity skew that single-factor Heston cannot reach without driving  $\eta$  to its cap, so it speaks directly to the degeneracy and Feller findings above. It does not, however, fit the routine of this paper. The dynamics are non-Markovian, so the finite-dimensional box, the seed grid, and the Feller quantity have no direct analogue. There is no QuantLib engine for it. The fastest tractable variant, rough Heston, prices through a fractional Riccati equation [El Euch and Rosenbaum, 2019] that is far heavier per evaluation than the closed-form Heston integral, and rough Bergomi [Bayer et al., 2016] prices by Monte Carlo. Multi-start calibration over thousands of days is then out of reach without a surrogate pricer. There is also a tension with the present surface: the rough skew is sharpest at the shortest maturities, exactly the tenors the pre-filter drops at  $\tau_{\min} = 14$  days (17). A fair test would re-admit the short end, build a rough-Heston pricer or a learned surrogate, and run on a pilot window rather than the full sample. We leave it to separate work.

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### Appendix A. Example option prices

Heston model prices produced by the market<sup>1</sup> calibrated pricing operator:

$$C_H(\Phi^*; S, K, \tau, w) \quad (10) \quad (36).$$

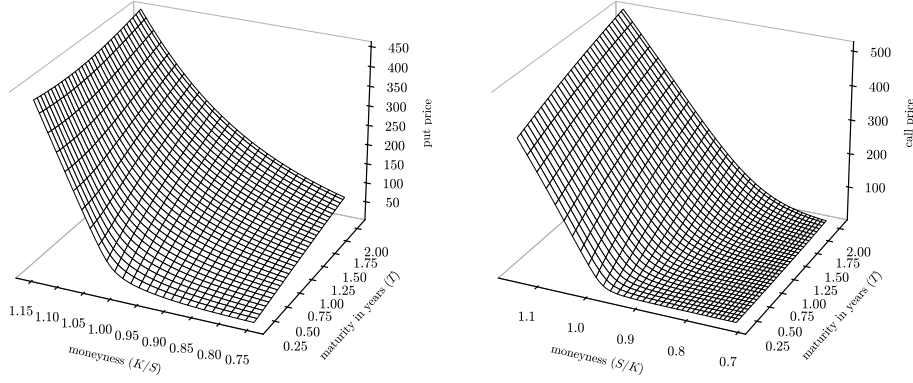


FIGURE 1. Heston option prices for  $S_{\text{ref}} = 2939.9736$  on May 03, 2019 with  $\Phi^* = (0.0431, 3.8796, 1.77, -0.6903, 0.0131)$ : puts wing (left) and calls wing (right).

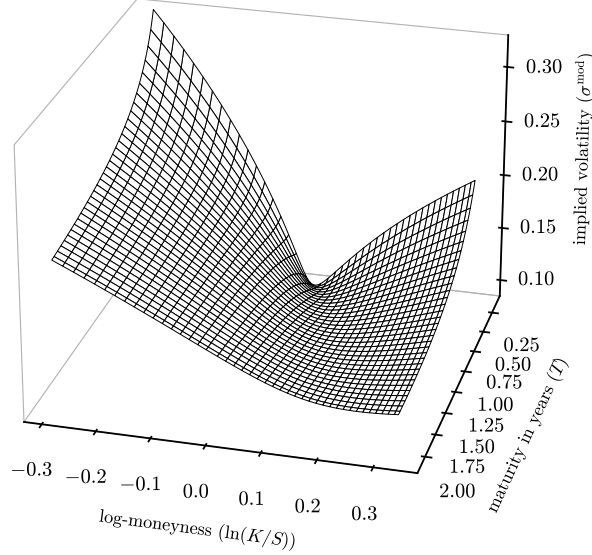


FIGURE 2. *Out of the money* implied volatilities from Figure 1

<sup>1</sup>The fit uses 1057 calibration cells across 20 maturities and 212 strikes, backed by 289,441 contracts of traded volume. The reference spot was  $S_{\text{ref}} = 2939.9736$ , priced under a risk-free rate of 2.53% and a dividend rate of 1.88%. The intraday spot range was 0.87%. Fit quality is 0.37 vol points of implied-volatility RMSE, with a relative-price RMSE of 0.0872. The Feller condition violates at this calibration, with  $2\kappa\theta - \eta^2 = -2.799$ .

**Bates model prices** produced by the market<sup>2</sup> calibrated pricing operator:

$$C_{\text{Bates}}(\Theta^*; S, K, \tau, w) \quad (14) \quad (36).$$

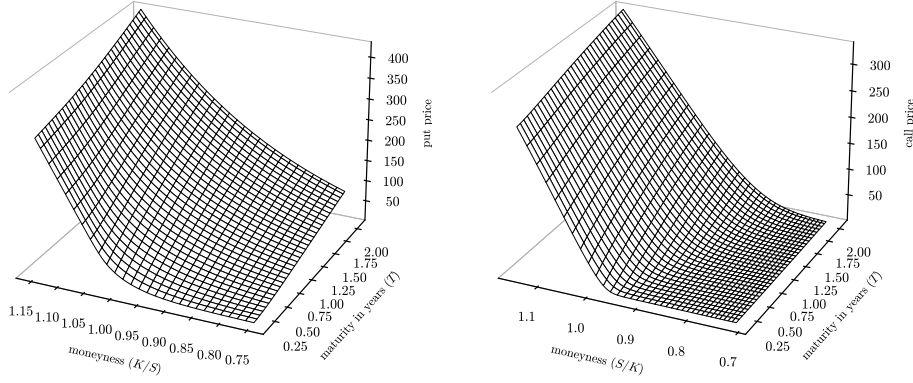


FIGURE 3. Bates option prices for  $S_{\text{ref}} = 2120.0581$  on May 15, 2015 with  $\Theta^* = (0.0582, 1.1934, 0.5949, -0.86, 0.0111, 0.0397, -0.3142, 0.3879)$ : puts wing (left) and calls wing (right).

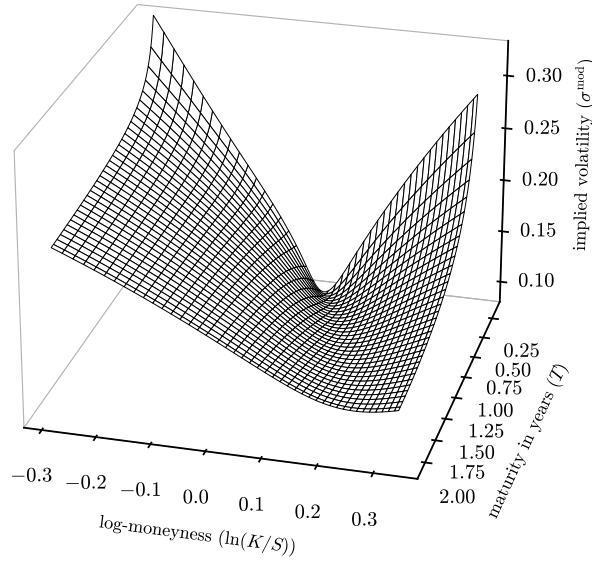


FIGURE 4. *Out of the money* implied volatilities from Figure 3

<sup>2</sup>The fit uses 653 calibration cells across 18 maturities and 118 strikes, backed by 267,772 contracts of traded volume. The reference spot was  $S_{\text{ref}} = 2120.0581$ , priced under a risk-free rate of 0.19% and a dividend rate of 1.97%. The intraday spot range was 0.33%. Fit quality is 0.27 vol points of implied-volatility RMSE, with a relative-price RMSE of 0.1063. The Feller condition violates at this calibration, with  $2\kappa\theta - \eta^2 = -0.215$ .

## Appendix B. Example fits for the Heston model

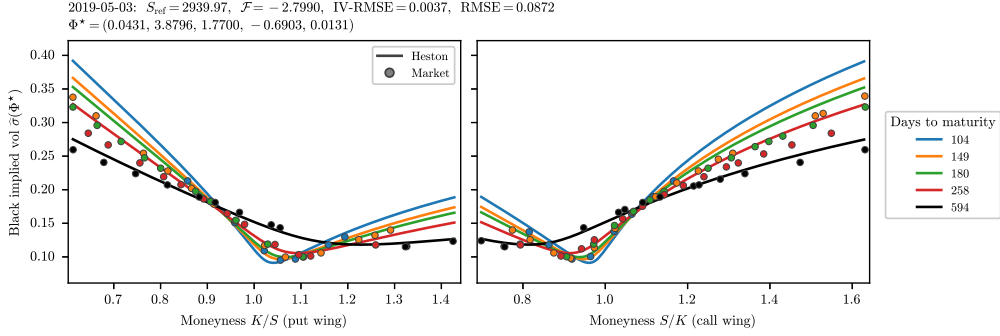


FIGURE 5. Heston implied-volatility smiles for May 03, 2019: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

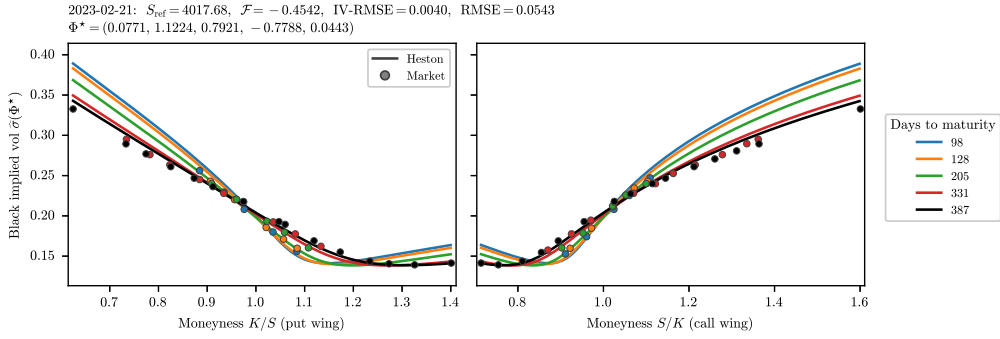


FIGURE 6. Heston implied-volatility smiles for February 21, 2023: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

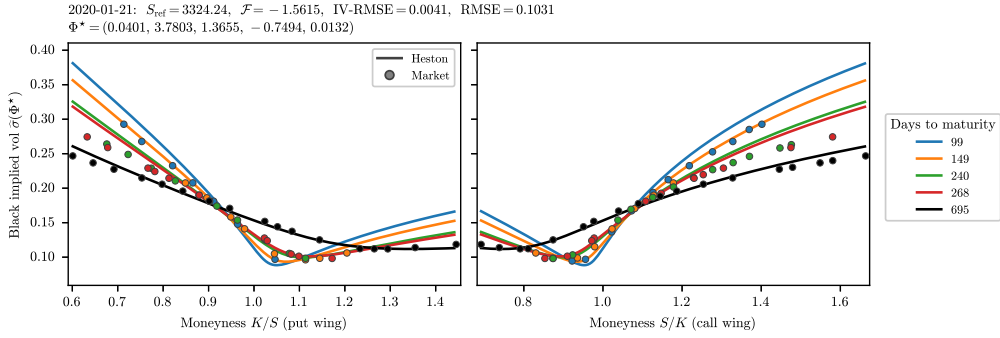


FIGURE 7. Heston implied-volatility smiles for January 21, 2020: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

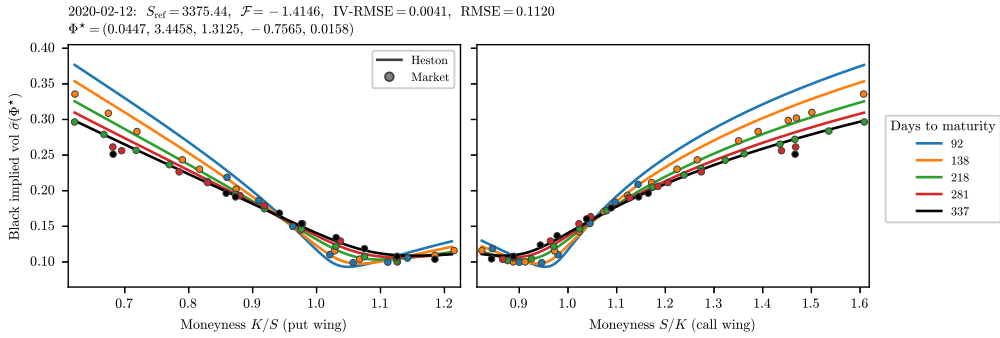


FIGURE 8. Heston implied-volatility smiles for February 12, 2020: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

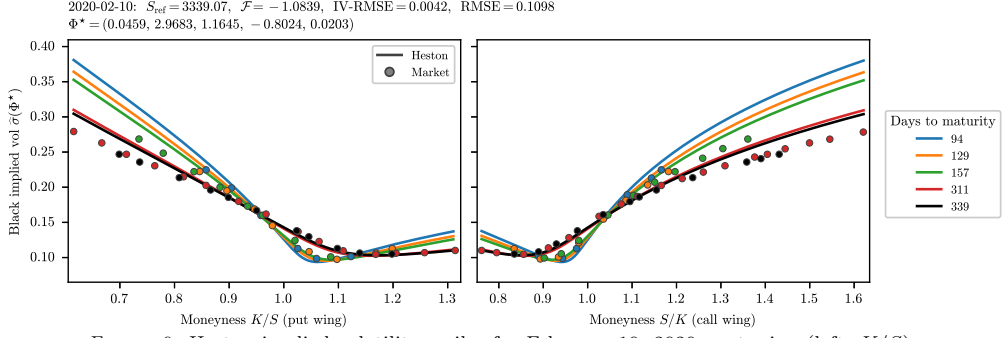


FIGURE 9. Heston implied-volatility smiles for February 10, 2020: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

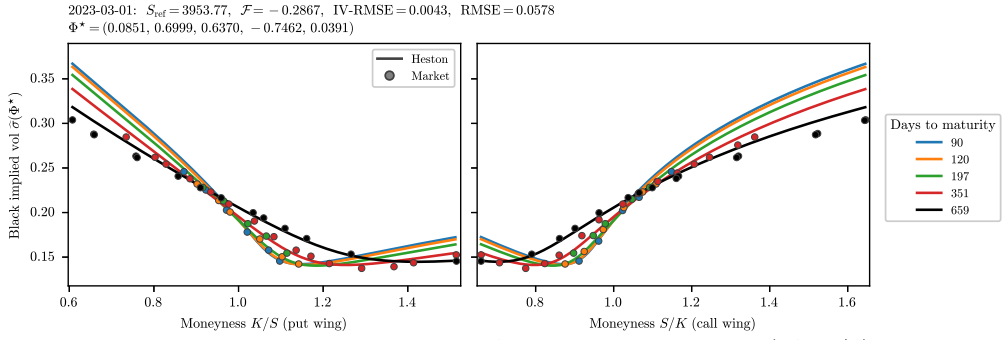


FIGURE 10. Heston implied-volatility smiles for March 01, 2023: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

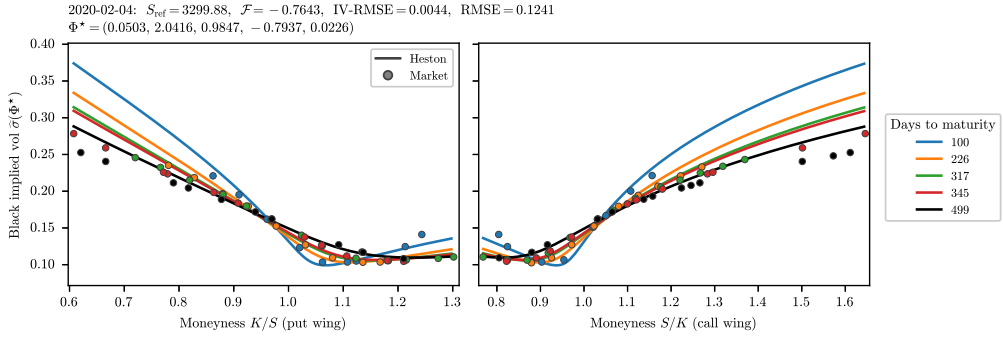


FIGURE 11. Heston implied-volatility smiles for February 04, 2020: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

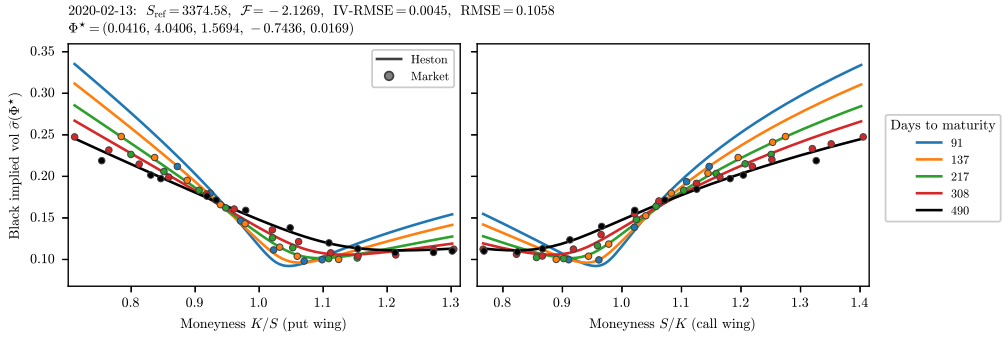


FIGURE 12. Heston implied-volatility smiles for February 13, 2020: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

### Appendix C. Example fits for the Bates model

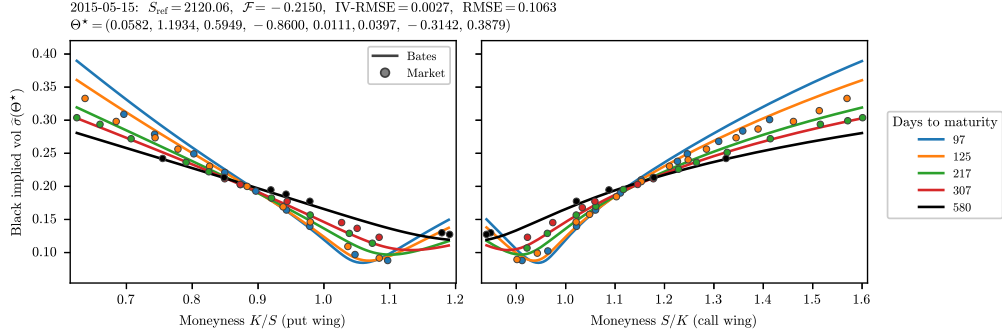


FIGURE 13. Bates implied-volatility smiles for May 15, 2015: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

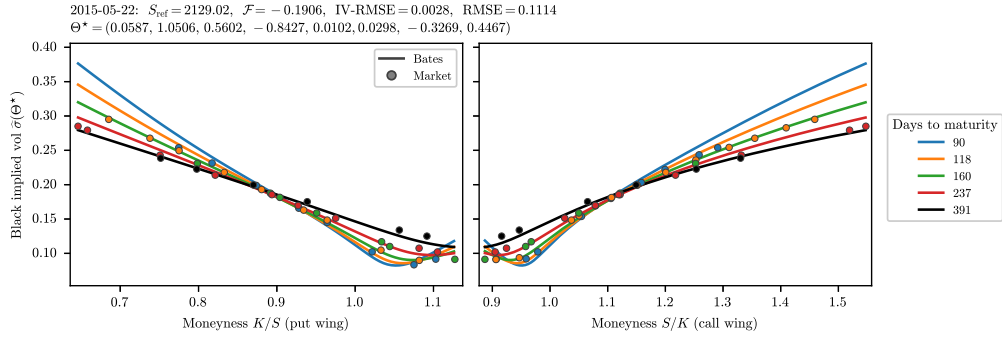


FIGURE 14. Bates implied-volatility smiles for May 22, 2015: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

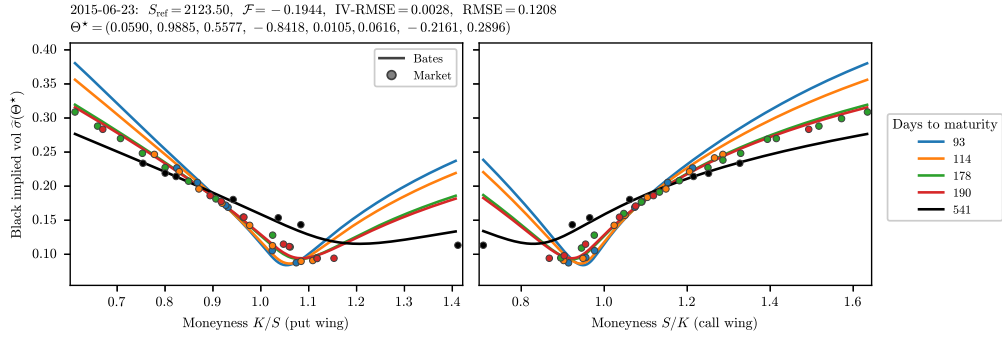


FIGURE 15. Bates implied-volatility smiles for June 23, 2015: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

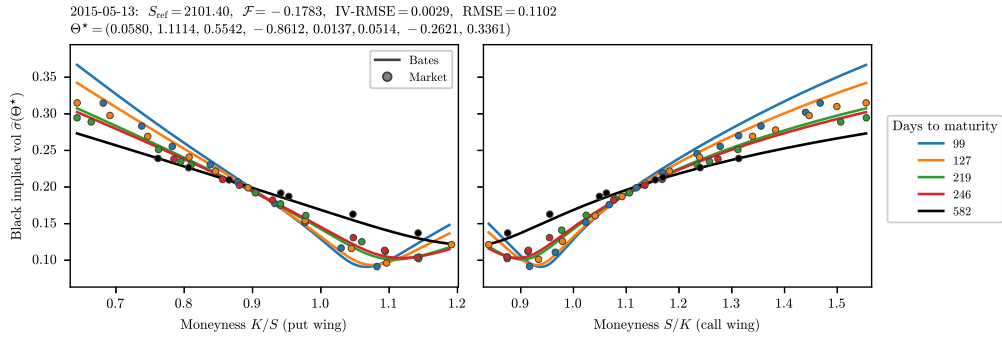


FIGURE 16. Bates implied-volatility smiles for May 13, 2015: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

2015-06-12:  $S_{ref} = 2096.05$ ,  $\mathcal{F} = -0.1400$ , IV-RMSE = 0.0029, RMSE = 0.1127  
 $\Theta^* = (0.0584, 0.8090, 0.4842, -0.8799, 0.0140, 0.0612, -0.2576, 0.3243)$

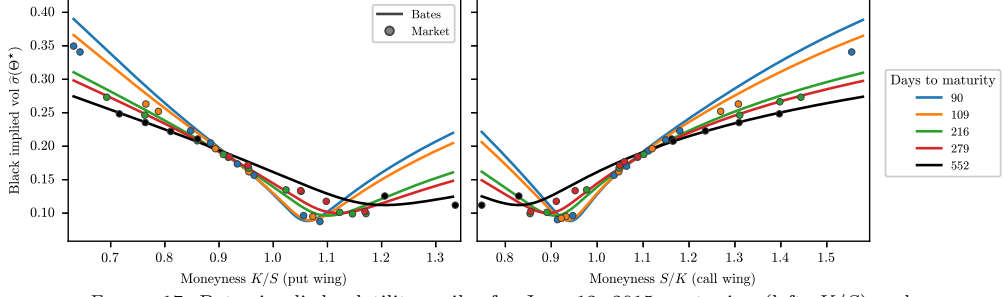


FIGURE 17. Bates implied-volatility smiles for June 12, 2015: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

2015-05-18:  $S_{ref} = 2126.56$ ,  $\mathcal{F} = -0.1664$ , IV-RMSE = 0.0029, RMSE = 0.1211  
 $\Theta^* = (0.0641, 0.8387, 0.5234, -0.8639, 0.0114, 0.0455, -0.3209, 0.3929)$

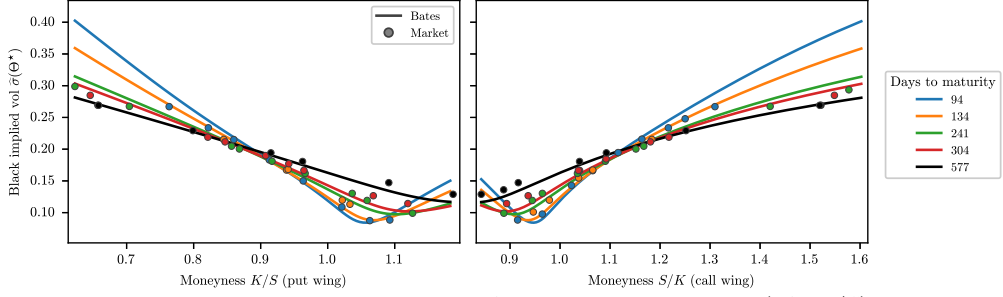


FIGURE 18. Bates implied-volatility smiles for May 18, 2015: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

2015-06-11:  $S_{ref} = 2110.15$ ,  $\mathcal{F} = -0.1655$ , IV-RMSE = 0.0030, RMSE = 0.1491  
 $\Theta^* = (0.0572, 0.9489, 0.5235, -0.8602, 0.0121, 0.0486, -0.2672, 0.3527)$

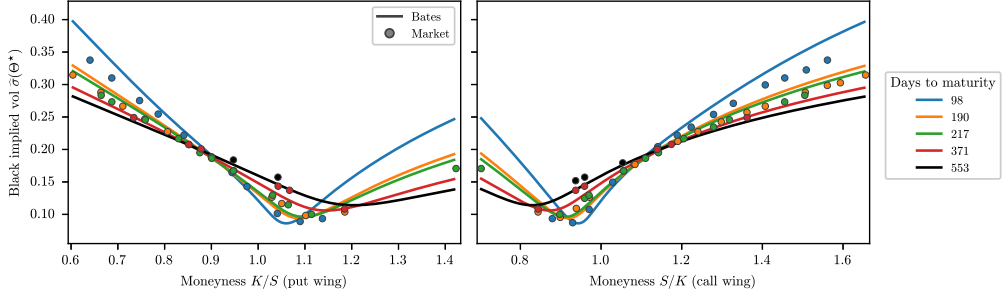


FIGURE 19. Bates implied-volatility smiles for June 11, 2015: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

2015-05-19:  $S_{ref} = 2128.74$ ,  $\mathcal{F} = -0.2359$ , IV-RMSE = 0.0030, RMSE = 0.1210  
 $\Theta^* = (0.0670, 0.8643, 0.5930, -0.8343, 0.0113, 0.0486, -0.2633, 0.3286)$

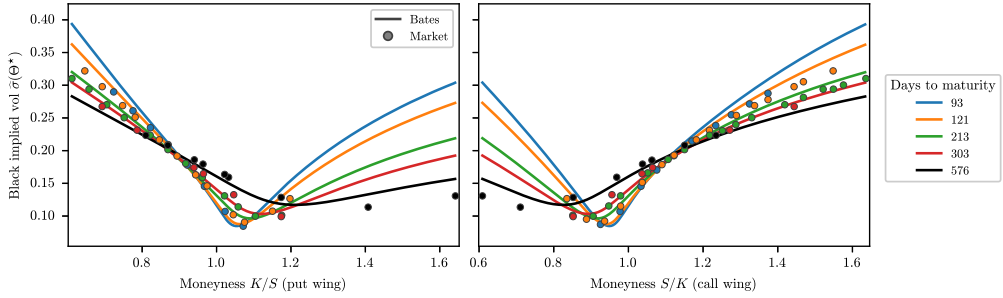


FIGURE 20. Bates implied-volatility smiles for May 19, 2015: put wing (left,  $K/S$ ) and call wing (right,  $S/K$ ), with market trades scattered.

### Appendix D. Algorithm

Algorithm 1 states the per-day routine in the active vector  $\Psi$ ; the driver applies it to every trading day and accumulates accepted rows into a single file (§7). Heston is the default ( $\Psi = \Phi$ ). The Bates fit is the same algorithm with  $\Psi = \Theta$ : the engine is the **BatesEngine** and the box  $\mathcal{B}$  is the dim  $\Theta$ -dimensional one, while the pegging test (35) still ranges over the dim  $\Phi$  diffusive coordinates only.

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**Algorithm 1** Per-day calibration
 

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**Require:** day  $d$  of OTM trades; rates  $r(d), g(d)$ ; constants of Table 1

- 1:  $\mathcal{D} \leftarrow \{i : \sigma_i > 0, \tau_i \geq \tau_{\min}\}$  (pre-filter (17))
- 2: **if**  $r(d)$  or  $g(d)$  undefined on/before  $d$  **then**
- 3:   **return**  $\emptyset$  (no rate)
- 4: **end if**
- 5:  $S_{\text{ref}} \leftarrow (\sum q_i S_i) / (\sum q_i)$ ;  
 $R \leftarrow \max_i S_i / \min_i S_i - 1$ ;  
 $\text{HIGH\_MOVE} \leftarrow \mathbf{1}\{R > R_{\max}\}$
- 6:  $K_i^* \leftarrow \Delta \text{round}(m_i S_{\text{ref}} / \Delta)$  with  $m_i = K_i / S_i$  (re-strike (20))
- 7:  $\mathcal{T} \leftarrow \text{top-}\bar{N}_t$  admissible maturities by volume ( $\geq \underline{n}_k$  strikes per wing);  
 build  $\mathcal{K}_\tau$  (maximum  $\bar{n}_k$  per wing);  
 pivot to  $\{\hat{\sigma}_{k,\tau}\}$ , cells  $\mathcal{N}$
- 8: **if** coverage (25) fails **then**
- 9:   clear stale diagnostics; **return**  $\emptyset$
- 10: **end if**
- 11:  $\hat{v} \leftarrow \text{clip}(\text{median}(\hat{\sigma})^2, [10^{-3}, 0.25])$ ; build seeds  $\mathcal{S}$  (31) (32)
- 12: **for**  $s \in \mathcal{S}$  **do**
- 13:    $\hat{\Psi}_s \leftarrow \text{LM minimiser of } \sum_n (\varepsilon_n^{\text{obj}})^2 \text{ on } \mathcal{B} \text{ from } s$  (33)
- 14:   compute  $J_{\text{iv}}(\hat{\Psi}_s)$  (30)
- 15: **end for**
- 16:  $\Psi^* \leftarrow \arg \min_s J_{\text{iv}}(\hat{\Psi}_s)$  (select (34))
- 17: **if**  $J_{\text{iv}}(\Psi^*) \leq \varepsilon_{\text{iv}}$  **and**  $\neg \text{PEGGED}(\Psi^*)$  **then**
- 18:   write repriced diagnostics; **return**  $\text{row}(d, S_{\text{ref}}, r, g, \Psi^*, \mathcal{F}, J_{\text{iv}}, J_{\text{px}}, \dots)$
- 19: **else**
- 20:   clear stale diagnostics; **return**  $\emptyset$  (rejected)
- 21: **end if**

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